



MTH401 Final TERM Solved Subjective

Dynamics (Virtual University of Pakistan)



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MTH401 FINAL TERM SOLVED SUBJECTIVE
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Q1. Write down the system of differential equations

$$\frac{dx}{dt} = 6x + y + 6t, \quad \frac{dy}{dt} = 4x + 3y - 10t + 4$$

in form of $X' = AX + F(t)$

Solution:

$$X' = \begin{pmatrix} 6 & 1 \\ 4 & 3 \end{pmatrix} X + \begin{pmatrix} 6t \\ -10t + 4 \end{pmatrix}$$

$$\frac{dy}{dt} = x, \quad \frac{dx}{dt} = y$$

Q2. Solve the system of differential equations by systematic elimination.

Solution:

$$\frac{dy}{dt} = x \Rightarrow Dy - x = 0 \quad \dots\dots\dots(i)$$

$$\frac{dx}{dt} = y \Rightarrow -y + Dx = 0 \quad \dots\dots\dots(ii)$$

Operate (ii) by D , we get

$$-Dy + D^2x = 0 \quad \dots\dots\dots(iii)$$

Add (i) and (iii), we get

$$Dy - x = 0$$

$$\underline{-Dy + D^2x = 0}$$

$$D^2x - x = 0$$

$$(D^2 - 1)x = 0$$

Auxiliary equation is $m^2 - 1 = 0$

$$m = \pm 1$$

$$\boxed{x(t) = c_1 e^t + c_2 e^{-t}}$$

Put this in (i), we get

$$Dy - [c_1 e^t + c_2 e^{-t}] = 0$$

$$Dy = c_1 e^t + c_2 e^{-t}$$

Integrate both sides, we get

$$\boxed{y(t) = c_1 e^t - c_2 e^{-t}}$$

Q3. Find a series solution for the differential equation $y'' + y = 0$ about $x_0 = 0$ such that

$$\alpha_{n+2} = -\frac{\alpha_n}{(n+2)(n+1)} \quad n = 0, 1, 2, \dots \quad \text{and} \quad y(x) = \sum_{n=0}^{\infty} a_n x^n$$

Solution:

$$a_{n+2} = -\frac{a_n}{(n+2)(n+1)}; \quad n = 0, 1, 2, \dots$$

$$\text{For } n = 0, a_2 = -\frac{a_0}{(0+2)(0+1)} = -\frac{a_0}{2}$$

$$\text{For } n = 1, a_3 = -\frac{a_1}{(1+2)(1+1)} = -\frac{a_1}{6}$$

$$\text{For } n = 2, a_4 = -\frac{a_2}{(2+2)(2+1)} = -\frac{a_2}{12} = -\frac{1}{12}\left(-\frac{a_0}{2}\right) = \frac{a_0}{24}$$

$$\text{For } n = 3, a_5 = -\frac{a_3}{(3+2)(3+1)} = -\frac{a_3}{20} = -\frac{1}{20}\left(-\frac{a_1}{6}\right) = \frac{a_1}{120}$$

$$y(x) = a_0 + a_1x + a_2x^2 + a_3x^3 + a_4x^4 + a_5x^5 + \dots$$

$$y(x) = a_0 + a_1x + \left(-\frac{a_0}{2}\right)x^2 + \left(-\frac{a_1}{6}\right)x^3 + \left(\frac{a_0}{24}\right)x^4 + \left(\frac{a_1}{120}\right)x^5 + \dots$$

$$y(x) = a_0\left(1 - \frac{1}{2}x^2 + \frac{1}{24}x^4 + \dots\right) + a_1\left(x - \frac{1}{6}x^3 + \frac{1}{120}x^5 + \dots\right)$$

$$X = \frac{4}{3}\cos 3t - \frac{5}{3}\sin 3t$$

$$X = A \sin(\omega t + \phi)$$

Q4. Write solution

in the form

$$A = \sqrt{\left(\frac{4}{3}\right)^2 + \left(-\frac{5}{3}\right)^2} = \frac{\sqrt{41}}{3}$$

$$\phi = \tan^{-1}\left(\frac{4/3}{-5/3}\right) = 0.6747 \text{ radians}$$

$$x(t) = \frac{\sqrt{41}}{3} \sin(3t + 0.6747)$$

If the equation

$$M(x, y)dx + N(x, y)dy = 0$$

Q5. is not exact.

Case 1:

When $\exists$ an integrating factor $u(y)$, a function of y only. This happens if the expression

$$\frac{\frac{\partial N}{\partial x} - \frac{\partial M}{\partial y}}{M}$$

is a function of y

Case2:

If the given equation is homogeneous and

$$xM + yN \neq 0$$

Then find the integrating factor in both cases.

Solution:

$$u = \frac{1}{xM + yN}$$

Q6. Find eigen vector of

$$\begin{pmatrix} -3 & 1 \\ 2 & -4 \end{pmatrix}$$

A= ,corresponding eigen value $\lambda = -2$.

$$\left(\begin{array}{cc|c} -3 - (-2) & 1 & 0 \\ 2 & -4 - (-2) & 0 \end{array} \right)$$

$$\left(\begin{array}{cc|c} -1 & 1 & 0 \\ 2 & -2 & 0 \end{array} \right)$$

Add two times row 1 in row 2

$$\left(\begin{array}{cc|c} -1 & 1 & 0 \\ 0 & 0 & 0 \end{array} \right)$$

$$-k_1 + k_2 = 0$$

$$k_1 = k_2$$

Choosing $k_2 = 1$, we get $k_1 = 1$

therefore, eigen vector is $V = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

Q8. Under which conditions linear independence of the solutions for the differential equation $y'' + P(x)y' + Q(x)y = 0$ (1) is guaranteed?

Solution:

Linear independence is guaranteed in case when the Wronskian of the two solutions is not equal to zero.

Q9. Write the homogenous system of differential equations

$$\begin{aligned}2 \frac{dx}{dt} - 5x + \frac{dy}{dt} &= 5e^t \\ \frac{dx}{dt} - x + \frac{dy}{dt} &= e^t\end{aligned}$$

lec 36 example 1

in decoupled form.

$$2 \frac{dx}{dt} - 5x + \frac{dy}{dt} = 5e^t$$

$$\frac{dx}{dt} - x + \frac{dy}{dt} = e^t$$

$$(2D - 5)x + Dy = 5e^t$$

$$(D - 1)x + Dy = e^t$$

Determinants are $\begin{vmatrix} 2D-5 & D \\ D-1 & D \end{vmatrix}, \begin{vmatrix} 5e^t & D \\ e^t & D \end{vmatrix}, \begin{vmatrix} 2D-5 & 5e^t \\ D-1 & e^t \end{vmatrix}$

Therefore, in decoupled form, we get

$$\begin{vmatrix} 2D-5 & D \\ D-1 & D \end{vmatrix} x = \begin{vmatrix} 5e^t & D \\ e^t & D \end{vmatrix}$$

$$\begin{vmatrix} 2D-5 & D \\ D-1 & D \end{vmatrix} y = \begin{vmatrix} 2D-5 & 5e^t \\ D-1 & e^t \end{vmatrix}$$

Q10. When Frobenius' Theorem is used in Differential Equation

$$a_2(x)y'' + a_1(x)y' + a_0(x)y = 0$$

?

Solution:

When we have a regular singular point $x = x_0$, then we can find at least one series solution of the form $y = \sum_{n=0}^{\infty} c_n (x - x_0)^{n+r}$, where r is the constant that we will determine after solving the differential equation.

Q11. Find the eigenvalues of the following system

$$X' = \begin{pmatrix} 3 & -9 \\ 4 & -3 \end{pmatrix} X$$

Solution:

$$X' = \begin{pmatrix} 3 & -9 \\ 4 & -3 \end{pmatrix} X$$

$$A = \begin{pmatrix} 3 & -9 \\ 4 & -3 \end{pmatrix}$$

for eigen values, $|A - \lambda I| = 0$

$$\begin{vmatrix} 3 - \lambda & -9 \\ 4 & -3 - \lambda \end{vmatrix} = 0$$

$$(3 - \lambda)(-3 - \lambda) + 36 = 0$$

$$3(-3 - \lambda) - \lambda(-3 - \lambda) + 36 = 0$$

$$-9 - 3\lambda + 3\lambda + \lambda^2 + 36 = 0$$

$$-9 + \lambda^2 + 36 = 0$$

$$\lambda^2 + 27 = 0$$

$\lambda = \sqrt{27}i$ and $-\sqrt{27}i$ are the two complex eigen values

Q12. Define Legendre's polynomial of degree n

Solution:

Legendre polynomial is an n^{th} degree polynomial and it is given by the formula

$$P_n(x) = \frac{1}{2^n n!} \frac{d^n}{dx^n} (x^2 - 1)^n$$

Q13. What is the ordinary differential equation and give an example?

Solution:

A differential equation which only includes ordinary derivatives is known as ordinary differential equation. Some examples of ordinary differential equations include:

$$\frac{dy}{dx} = x^2 + y$$

$$\frac{dy}{dx} = (x^2 + 1)(y^2 + 1)$$

$$\frac{d^2y}{dx^2} + 2\frac{dy}{dx} + 3y = 0$$

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